DO NOW — RECALL

Lesson 132 · Hybrid Encryption: Protecting a Connectior

Name: _____ Class: _____ Date: _____

1. Which key verifies a signature produced using an owner's private key?

2. Which memory is typically volatile: RAM or ROM?

3. In a queue, what does FIFO stand for?

4. What is O(n) search complexity associated with?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The corresponding trusted public key.

2. Model answer

RAM.

3. Model answer

First in, first out.

4. Model answer

Linear/sequential search in the worst case.

By the end, you can…

• Client validates server identity and securely establishes shared key material.

• Both ends obtain secret symmetric traffic keys.

• Bulk application data travels encrypted under symmetric cipher.

Vocabulary — know these terms

session key — Temporary symmetric key for a connection/session.

hybrid encryption — Combines asymmetric key establishment with symmetric data encryption.

certificate — Credential binding identity to a public key, subject to trust validation.

confidentiality — Preventing unauthorised reading of information.

NEW LEARNING

Read, annotate and explain

Authenticate and agree

Client validates server identity and securely establishes shared key material.

Derive session key

Both ends obtain secret symmetric traffic keys.

Encrypt traffic

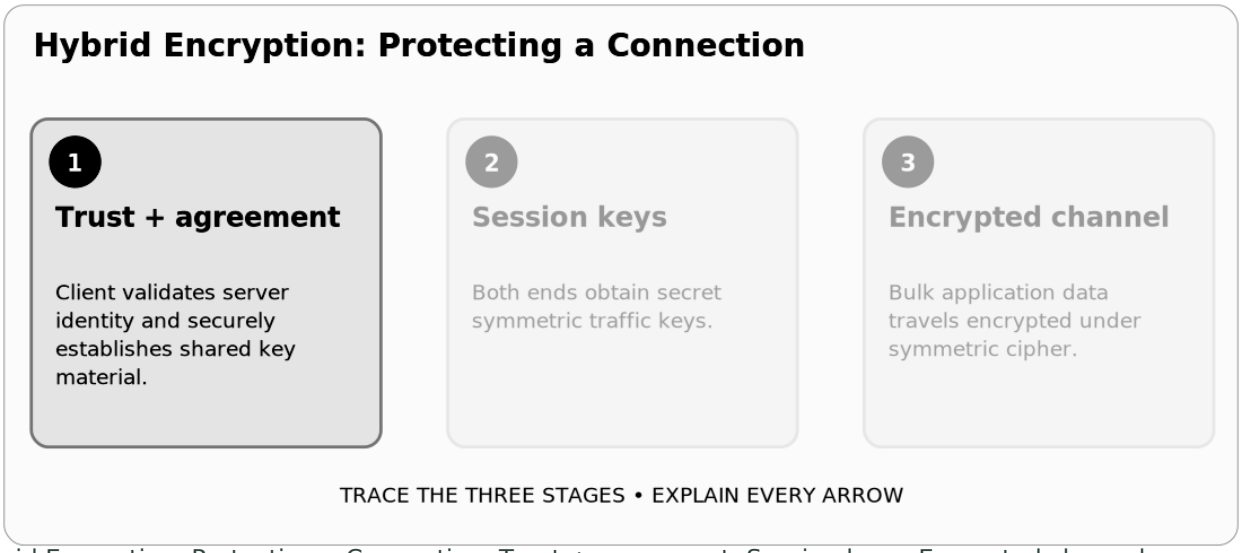
Bulk application data travels encrypted under symmetric cipher.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

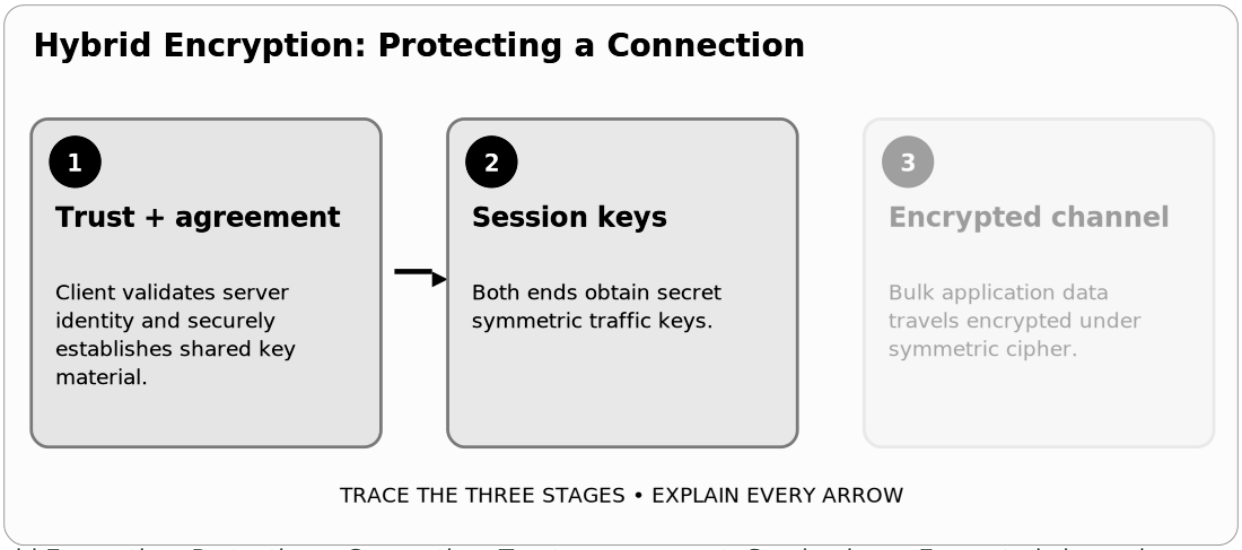
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Hybrid Encryption: Protecting a Connection: Trust + agreement; Session keys; Encrypted channel

New Learning

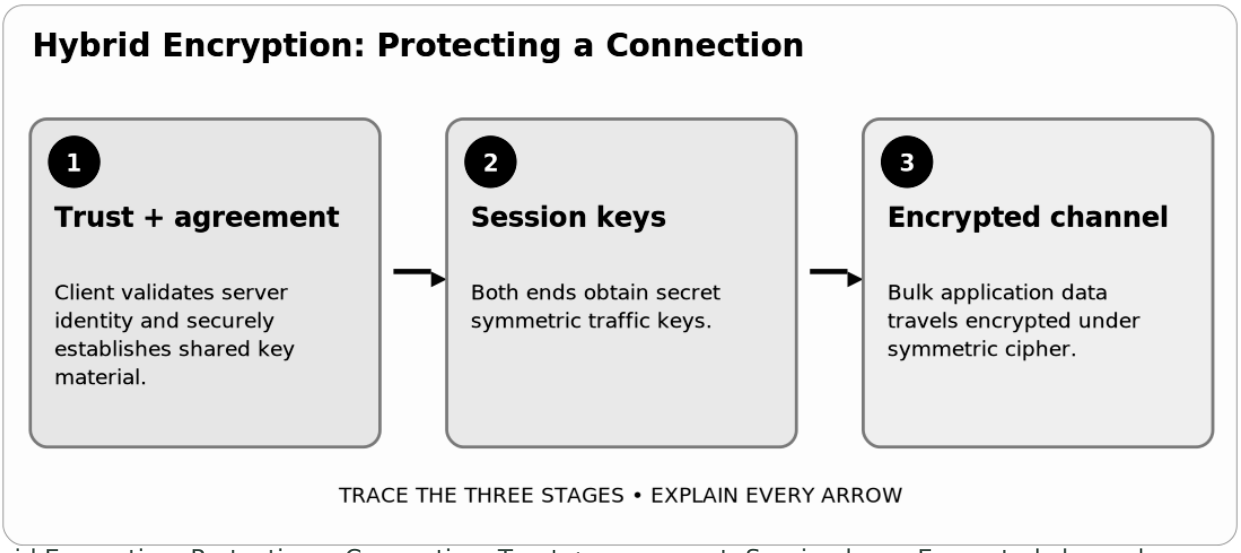


Hybrid Encryption: Protecting a Connection: Trust + agreement; Session keys; Encrypted channel

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Hybrid Encryption: Protecting a Connection: Trust + agreement; Session keys; Encrypted channel

I DO — WORKED EXAMPLE

Follow the worked method

Explain why a modern web connection does not encrypt every file byte directly with an asymmetric cipher.

Check server certificate

Predict the first step from the prompt.

Derive symmetric traffic key

Explain the intermediate result or decision.

Encrypt application data

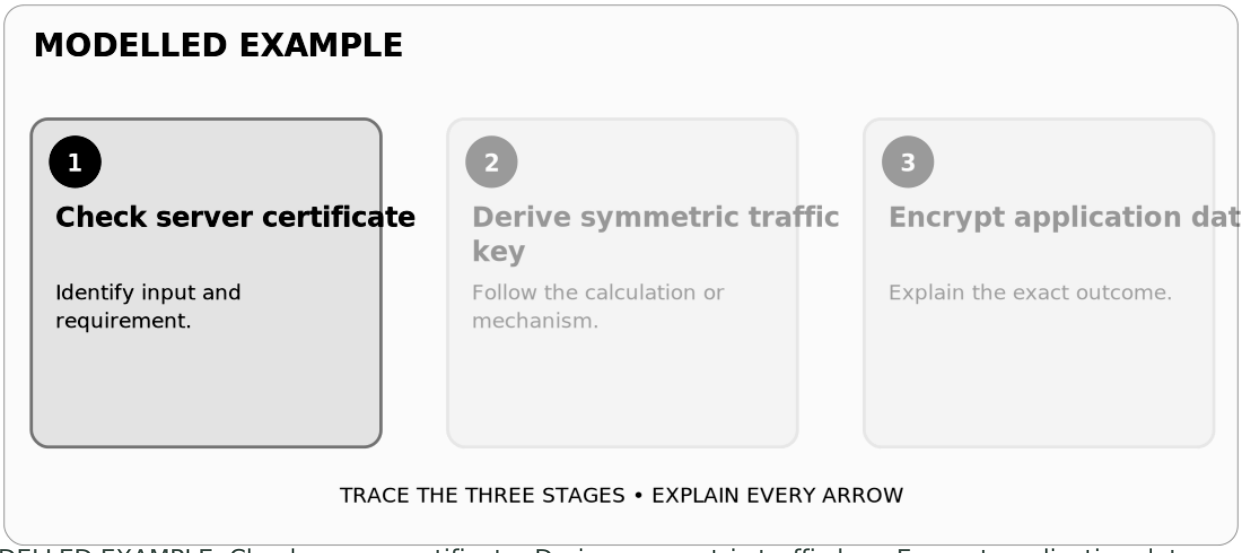
Modern secure connections typically authenticate with public-key credentials, establish shared session keys (often via ephemeral key agreement) and use efficient symmetric encryption for bulk traffic. Identity checks guard against impostors.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Explain why a modern web connection does not encrypt every file byte directly with an asymmetric cipher.

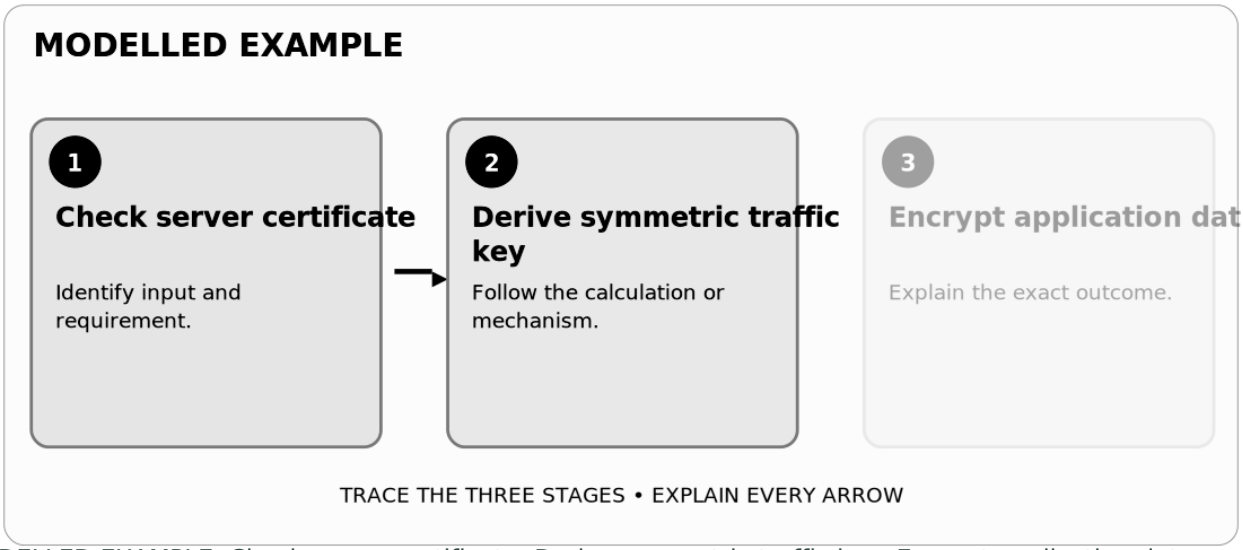
Predict the first step from the prompt.



MODELLED EXAMPLE: Check server certificate; Derive symmetric traffic key; Encrypt application data

Explain why a modern web connection does not encrypt every file byte directly with an asymmetric cipher.

Predict the first step from the prompt.



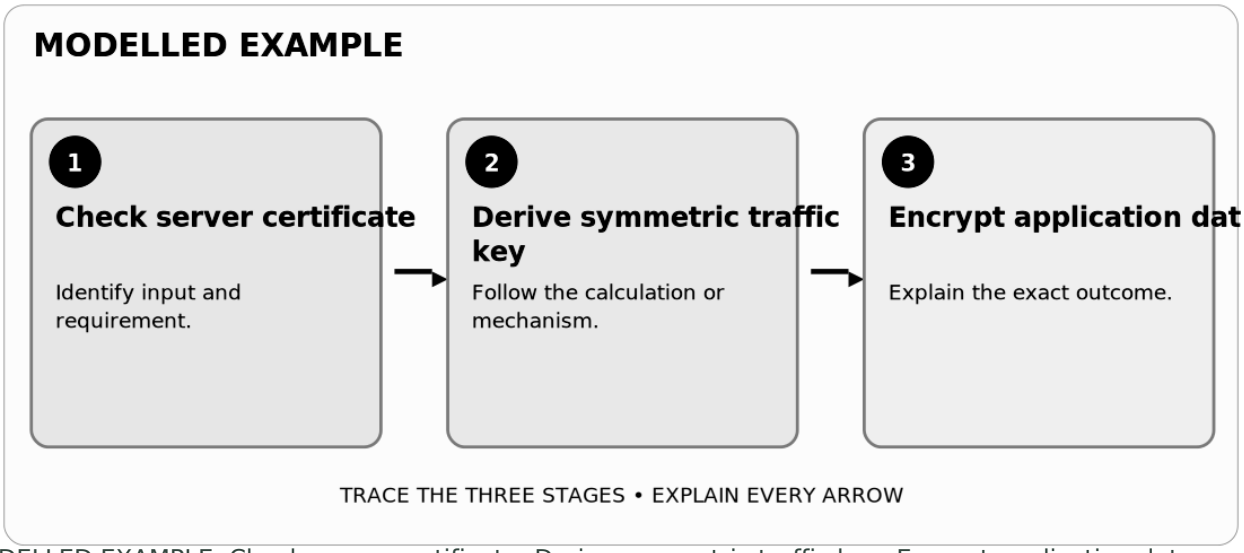
MODELLED EXAMPLE: Check server certificate; Derive symmetric traffic key; Encrypt application data

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Explain why a modern web connection does not encrypt every file byte directly with an asymmetric cipher.

Predict the first step from the prompt.



MODELLED EXAMPLE: Check server certificate; Derive symmetric traffic key; Encrypt application data

WE DO — GUIDED PRACTICE

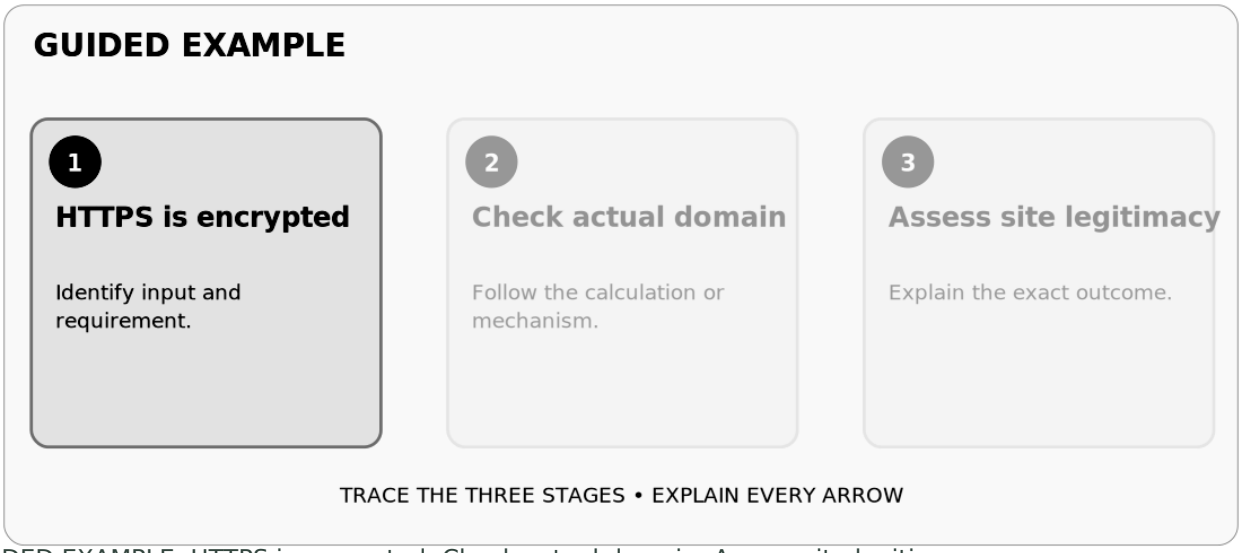
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil visits a fake website showing a padlock. Does encryption prove the website is trustworthy?

Predict the first step from the prompt.



GUIDED EXAMPLE: HTTPS is encrypted; Check actual domain; Assess site legitimacy

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: HTTPS is encrypted Predict the first step from the prompt.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Describe why hybrid cryptography uses asymmetric and symmetric methods together. [6]

Question 2

Explain what a session key is and why it is not the same as a website's public key. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate the assertion 'the padlock means any entered data are safe'. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

HTTPS guarantees the website operator is honest.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

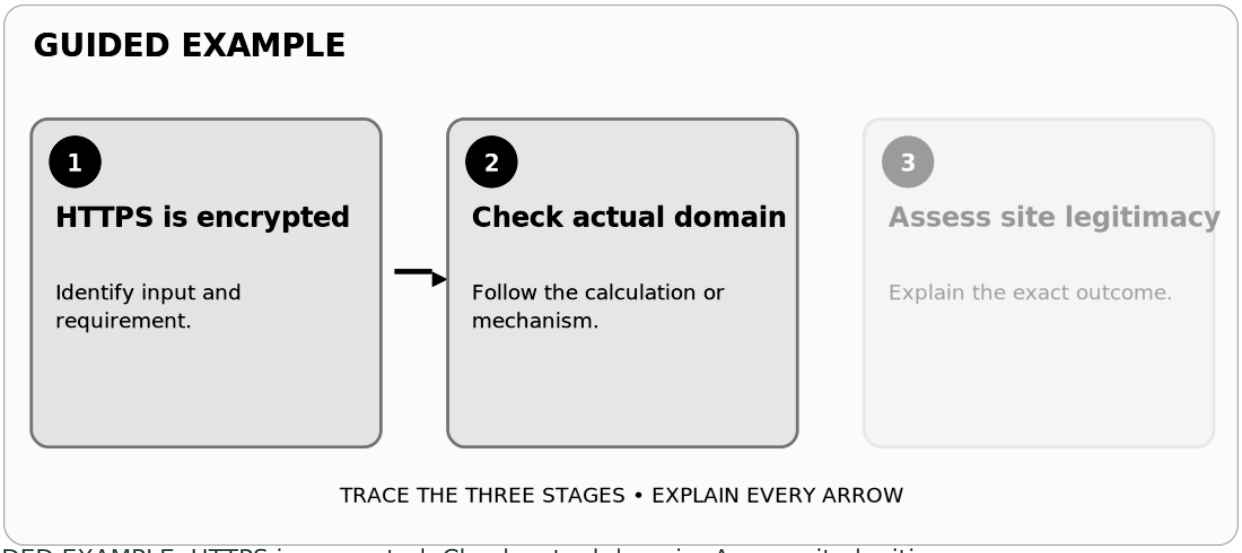
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil visits a fake website showing a padlock. Does encryption prove the website is trustworthy?

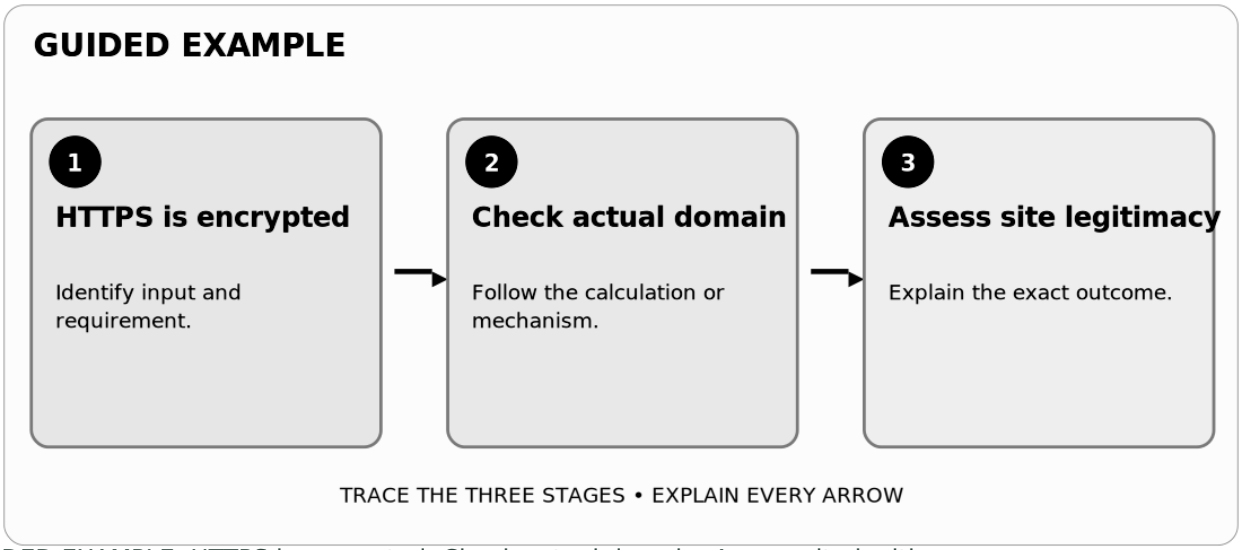
Predict the first step from the prompt.



GUIDED EXAMPLE: HTTPS is encrypted; Check actual domain; Assess site legitimacy

A pupil visits a fake website showing a padlock. Does encryption prove the website is trustworthy?

Predict the first step from the prompt.



GUIDED EXAMPLE: HTTPS is encrypted; Check actual domain; Assess site legitimacy

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check actual domain

Explain the intermediate result or decision.

Assess site legitimacy

No. HTTPS can establish an encrypted connection even with a harmful operator. Verify the domain and service, not just the presence of encrypted transport; a certificate proves control of an identity/domain to varying levels.

IMPROVE: Explain one correction to your own reasoning.